Flight-Control Incident Analysis
Autonomous Fixed-Wing UAV

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Abstract

This report examines the loss of control of a fixed-wing unmanned aircraft using the ArduPilot dataflash log. The analysis compares recorded radio-control inputs, commanded and measured attitude, aerodynamic state, altitude, and elevon outputs. The evidence indicates that a steep pitch-up command was followed by a progressive left-roll command that reached its recorded limit. The aircraft reached a high estimated angle of attack near the top of the climb, entered a steep descending left bank, and did not recover before ground impact. The recorded data do not, by themselves, demonstrate a mechanical control-surface failure.

1 Data and method

The principal data source was the ArduPilot dataflash file `Control surface issue.BIN`. The investigated interval was 64.0–78.2 s after flight-controller boot. All plots use the same time axis so that pilot commands, aircraft response, aerodynamic condition, and actuator demand can be compared directly. The relevant logged variables are listed in Table 1.

Table 1: Logged variables used in the incident analysis.

Message	Field(s)	Interpretation
`ATT`	`DesRoll`, `Roll`	Commanded and measured roll attitude
`ATT`	`DesPitch`, `Pitch`	Commanded and measured pitch attitude
`RCIN`	`C1`, `C2`, `C3`	Recorded roll, pitch, and throttle inputs
`AOA`	`AOA`	Estimated angle of attack
`ARSP`	`Airspeed`	Pitot-derived airspeed
`GPS`	`Spd`	GNSS-derived groundspeed
`BARO`	`Alt`	Barometric altitude relative to the start point
`RCOU`	`C1`, `C2`	Left- and right-elevon PWM outputs
`AETR`	`Ail`, `Elev`	Mixed aileron and elevator control demands

2 Results

2.1 Commanded and measured attitude

Figures 1 and 2 compare the commanded attitude with the aircraft's measured response. The commanded roll became progressively more negative and reached approximately -80 $^{\circ}$ at about 72 s. The measured roll subsequently exceeded -60 $^{\circ}$. During the attempted recovery, the commanded pitch approached 50 $^{\circ}$, while the measured pitch remained strongly negative. This divergence indicates that the requested recovery attitude could not be achieved within the remaining altitude and available aerodynamic control authority.

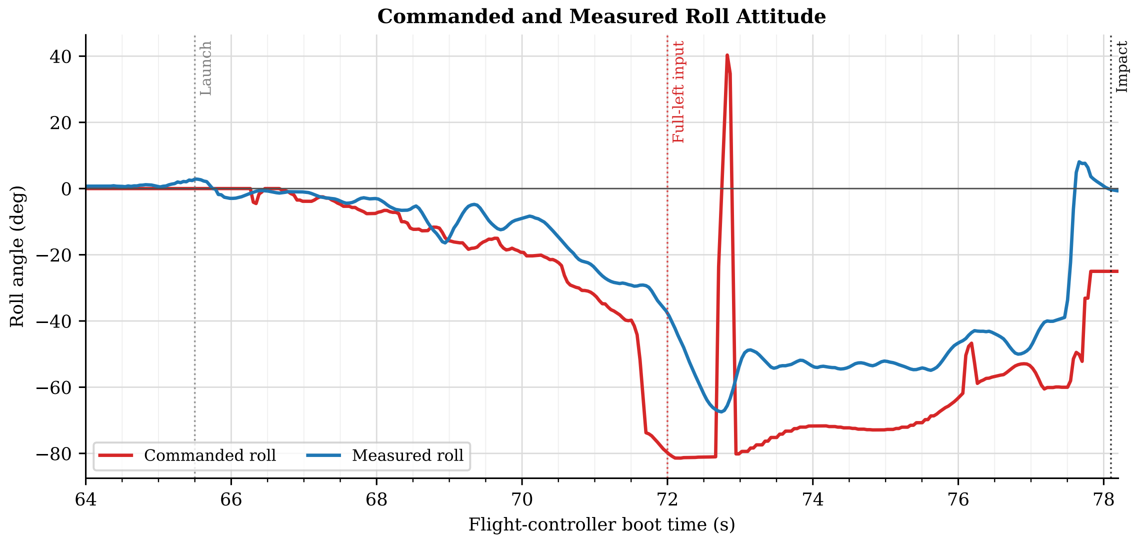

Figure 1: Commanded and measured roll attitude during the loss-of-control sequence. The commanded roll reached approximately $-80°$ after the recorded RC roll input reached its lower limit.

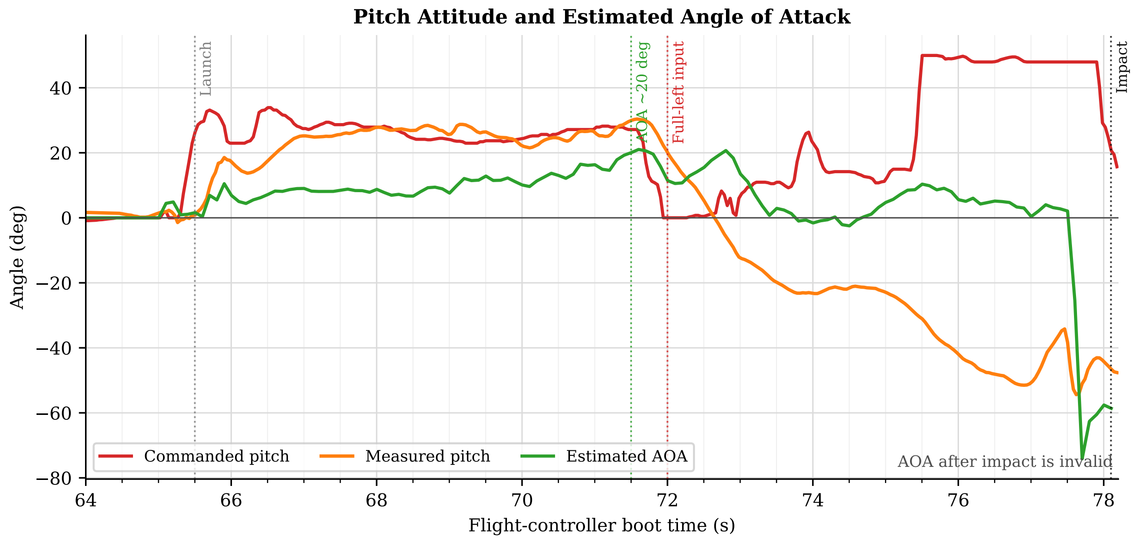

Figure 2: Commanded pitch, measured pitch and estimated angle of attack. The estimated angle of attack approached 20° near the top of the climb. Values recorded during impact are not aerodynamically valid.

2.2 Recorded radio-control inputs

Figure 3 presents the received PWM signals. Roll input `RCIN.C1` decreased gradually from its nominal centre of approximately 1500 μs before reaching exactly 1000 μs at about 72 s. It remained at that value through most of the descent. This establishes that the flight controller received a full-left roll command; however, the log alone cannot distinguish deliberate stick movement from a transmitter, gimbal, mixing, or receiver-channel fault. The pitch channel records pitch-up demand during both the initial climb and the later recovery attempt.

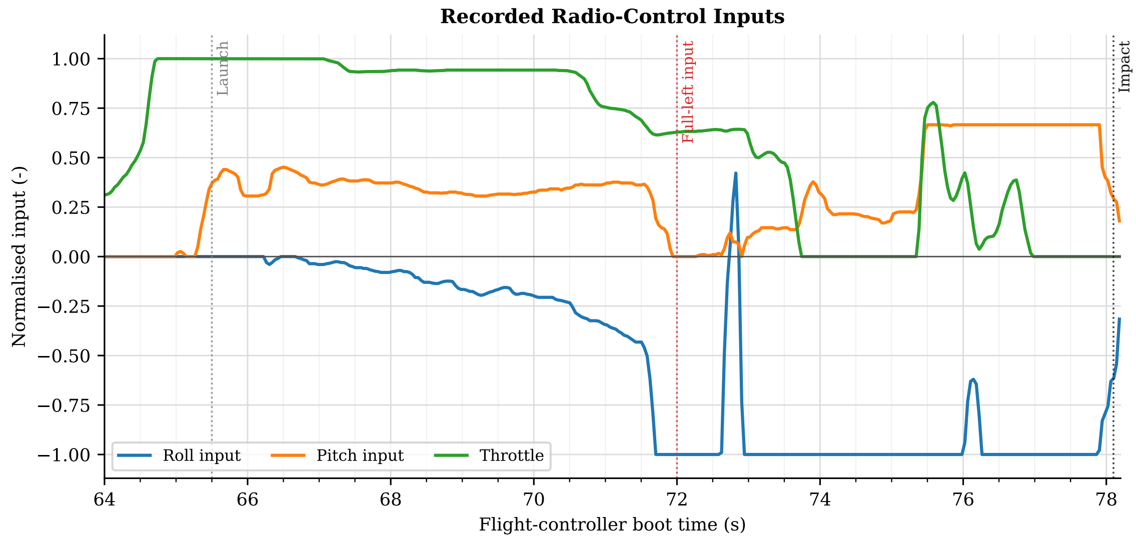

Figure 3: Normalised radio-control inputs received by the flight controller. The roll channel reached and predominantly remained at the full-left limit from approximately 72 s until impact. These signals are described as recorded RC inputs rather than confirmed pilot inputs.

2.3 Aerodynamic and energy state

Pitch attitude is not equivalent to angle of attack. To first order, the relationship is

$$\alpha \approx \theta - \gamma, \tag{1}$$

where α is angle of attack, θ is pitch attitude, and γ is flight-path angle. At approximately 71.5 s, the aircraft had a pitch attitude close to 30° and an estimated angle of attack close to 20°. Equation 1 therefore gives an approximate climb-path angle of only 10°. The difference suggests that the aircraft's nose was rising substantially faster than its flight path, consistent with a low-energy, high-angle-of-attack condition.

Figure 4 compares pitot-derived airspeed with GPS-derived groundspeed. The measured airspeed remained mostly above the configured 9 m/s minimum during the main manoeuvre; however, `ARSPD_USE=0`, so the pitot measurement was logged but was not selected for normal airspeed-based flight control.

The altitude in Figure 5 reached approximately 28 m before decreasing rapidly. The estimated angle of attack subsequently became unreliable during impact and low-speed tumbling; the final extreme negative value must not be interpreted as a valid aerodynamic measurement.

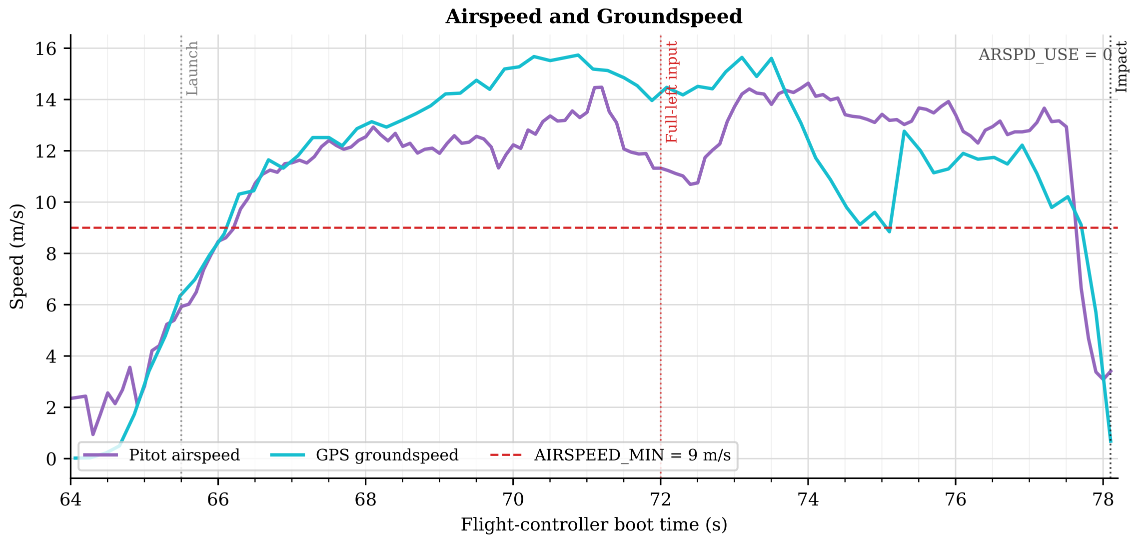

Figure 4: Pitot-derived airspeed and GPS-derived groundspeed. The configured minimum airspeed is included for reference; the recorded configuration had `ARSPD_USE=0`.

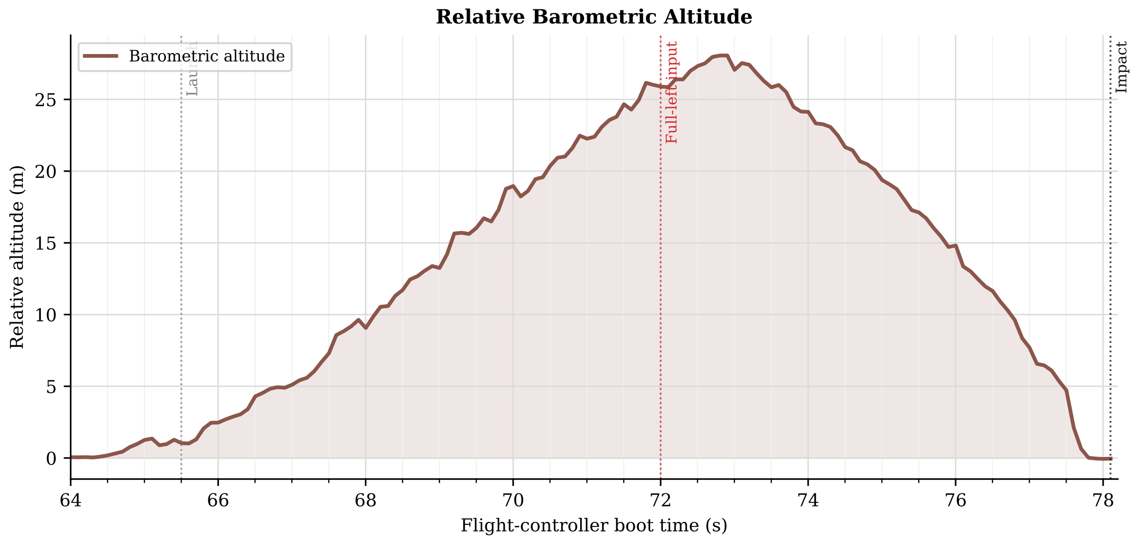

Figure 5: Barometric altitude relative to the start point. The aircraft reached approximately 28 m before entering the final descent.

2.4 Elevon allocation and saturation

Figure 6 compares the two elevon PWM outputs. The second output reached its configured upper limit of 2012 μs during the descending manoeuvre. This saturation occurred while large roll and pitch commands were being combined. It therefore demonstrates exhaustion of available commanded actuator range, but does not prove that either physical control surface failed to move. Surface position feedback or external video would be required to confirm a mechanical failure.

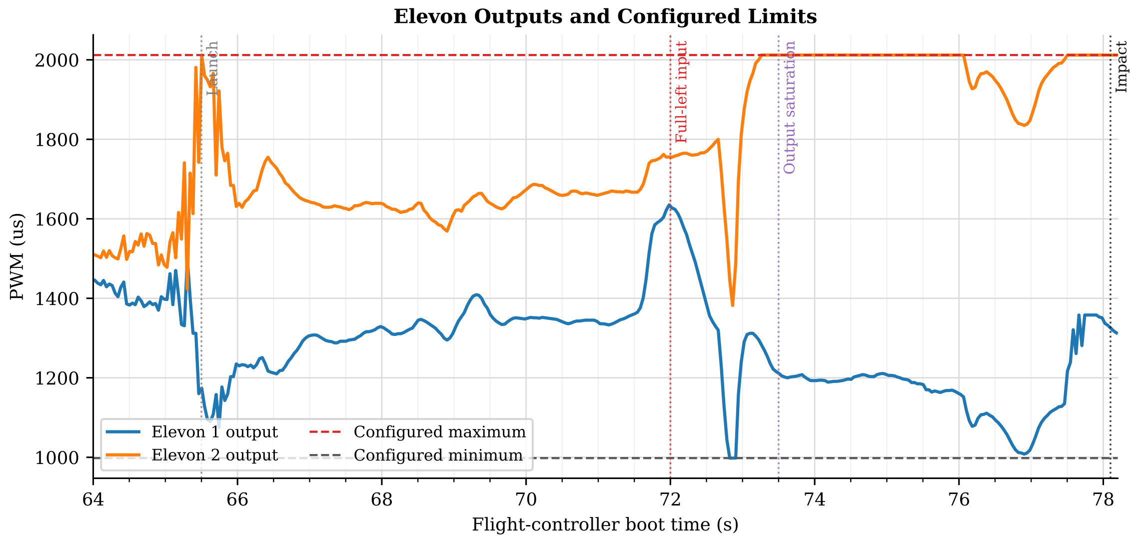

Figure 6: Left- and right-elevon PWM outputs with the configured limits. Elevon 2 reached its maximum output during the attempted recovery.

3 Event summary

Table 2: Principal events identified in the flight log.

Boot time (s)	Event
65.5	Pitch-up command and initial climb
71.5	Estimated AOA approaches 20 °; measured pitch approaches 30 °
72.0	Recorded roll input reaches full left; desired roll approaches -80 °
72.5	Measured roll exceeds approximately -60 ° near maximum altitude
73.5	Combined elevon output begins reaching its configured limit
75.5	Large pitch-up recovery command is recorded
78.1	Ground impact and throttle disarm

4 Relevant configuration

Table 3: Flight-control parameters relevant to the incident.

Parameter	Recorded value	Unit/state	Significance
`ROLL_LIMIT_DEG`	85	°	Permitted an exceptionally steep commanded bank
`PTCH_LIM_MAX_DEG`	60	°	Permitted an exceptionally steep pitch-up command
`PTCH_LIM_MIN_DEG`	-60	°	Permitted an exceptionally steep nose-down command
`STALL_PREVENTION`	1	enabled	Stall-prevention function was enabled
`ARSPD_USE`	0	disabled	Pitot airspeed was logged but not selected for normal flight control
`AIRSPEED_MIN`	9	m/s	Configured minimum airspeed
`MIXING_GAIN`	0.8	–	Elevon roll/pitch mixing gain

5 Conclusion

The most probable causal sequence was a steep pitch-up command, increasing angle of attack, a progressive left-roll command that reached its recorded limit, and a steep descending bank from which recovery was not completed before impact. The extreme configured attitude limits allowed FBWA mode to request attitudes that were unsuitable for a low-altitude prototype flight. The data show elevon output saturation during the recovery attempt, but they do not demonstrate an independent autopilot command or a mechanical control-surface failure.

Before further flight testing, the roll and pitch limits should be returned to conservative values; the source of the sustained 1000 μs roll input should be verified using the transmitter channel monitor; elevon direction, travel, mixing, and mechanical freedom should be checked under combined commands; and the airspeed system should be calibrated and configured appropriately.